# Scalable Blockchain-Based Traceability for Agricultural Imagery and Data Transformation

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## INTRODUCTION

- Ensuring **data traceability** in agriculture is essential for accurate monitoring, compliance, and informed decision-making. However, existing blockchain-based solutions face **scalability limitations** due to their reliance on rigid data structures. High transaction costs and inefficient data handling mechanisms further hinder their practical adoption in agriculture.
- Current blockchain frameworks store **static data references (e.g., IPFS hashes) on-chain**, limiting the ability to flexibly update and reuse data. Each transformation requires generating a new hash, **fragmenting the data traceability process and increasing storage redundancy**. Additionally, **current models do not support dynamic data interaction,** restricting seamless interoperability between data users and agricultural technology providers (ATPs).
- This study introduces a **scalable and flexible blockchain-based traceability framework** that enables **data reuse, transformation tracking, and controlled access without excessive on-chain storage.** Instead of logging all transformations on-chain, **a hashing ID update mechanism** ensures **continuous traceability while reducing scalability constraints.** This approach allows **authorized users to retrieve, modify, and re-upload datasets** while maintaining a verifiable history of all interactions. By integrating **dynamic data handling with blockchain security,** this model enhances **traceability, efficiency, and interoperability in agricultural data ecosystems.**

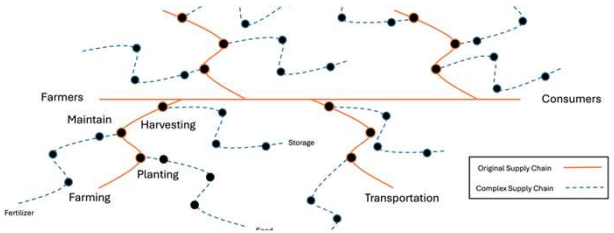
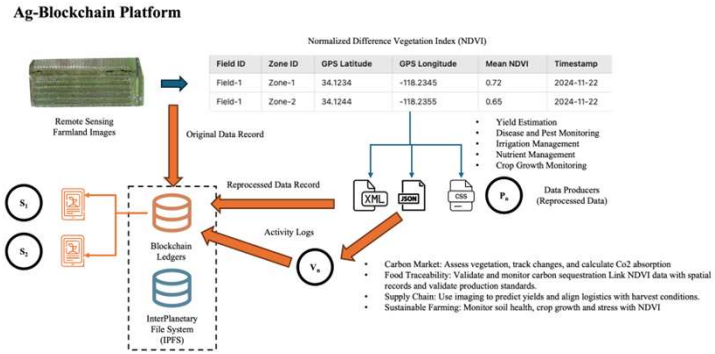



Fig. 1. Blockchain-based data branching and transformation for scalable agricultural data traceability.




| Field ID | Zone ID | GPS Latitude | GPS Longitude | Mean NDVI | Timestamp |
|---|---|---|---|---|---|
| Field-1 | Zone-1 | 34.1234 | -118.2345 | 0.72 | 2024-11-22 |
| Field-1 | Zone-2 | 34.1244 | -118.2355 | 0.65 | 2024-11-22 |

Fig. 2. Blockchain-based data storage and traceability system for agricultural imagery. The system integrates blockchain ledgers and the InterPlanetary File System (IPFS) to manage citrus drone imagery while maintaining traceability and efficient data handling.

## METHODS

- **Ag-Blockchain Prototype Development**
  A blockchain-based data tracking system was developed to monitor and record data activities using **drone imagery (TIFF)**. The system enables **data aggregation, transformation tracking, and selective logging**, ensuring traceability without excessive on-chain storage.

- **Smart Contract-Based Data Recording**
  A **smart contract** was implemented to **log all data-related activities**, including **uploads, access requests, transformations, downloads, and usage events**. Each interaction generates a unique **hashing ID**, which optimizes transaction speed and reduces storage overhead by preventing redundant on-chain storage of large datasets.

- **Data Handling & Scalability Solutions**
  To address scalability challenges in handling **large agricultural image files**, the system integrates an **Interplanetary File System (IPFS) transaction mechanism**, ensuring efficient storage while maintaining seamless access control and traceability.

- **Traceability & Dynamic Hashing Mechanism**
  Unlike conventional blockchain-based traceability models, this framework **assigns and updates dynamic hash IDs whenever data is transformed or accessed** (e.g., NDVI analysis, yield estimation). This approach ensures traceability without requiring redundant storage of intermediate datasets while enabling **efficient data retrieval and verification**.

- **Prototype Implementation & Evaluation**
  The prototype was tested using **citrus farm imagery from Immokalee, FL**, simulating various farm conditions and data processing scenarios. System performance was evaluated in terms of **traceability, efficiency, and scalability**, particularly focusing on **the impact of smart contract-based logging on transaction speed and storage optimization**.

## RESULTS

- **Blockchain-Based Data Tracking**
  All **user activities** related to **data uploads, updates, downloads, and transformations** were successfully recorded on the blockchain. Each transaction was logged with **timestamps, user IDs, and access roles**, which were converted into **secure hash values**, ensuring **tamper-proof data tracking and verification**.

- **Efficient Data Aggregation & Traceability**
  The system demonstrated **dynamic data aggregation and transformation tracking**, allowing datasets to be **reused, modified, and re-uploaded while maintaining full traceability**. Unlike conventional blockchain-IPFS models, the **hashing ID update mechanism** significantly reduced **redundant storage**, optimizing both **transaction speed and data retrieval efficiency**.

- **Scalability & Performance in Agricultural Image Processing**
  The framework was tested using **citrus drone imagery (TIFF),** applying predefined criteria such as **data type, size, timestamp, and geolocation.** Results showed that the **blockchain efficiently managed large agricultural datasets without excessive storage consumption**, confirming the system's ability to handle **real-world scalability demands while preserving data integrity.**

## FUTURE EFFORTS

- **Bridging Off-Chain Data Gaps with ZKP**
  Implement **Zero-Knowledge Proofs (ZKP)** to verify and track **off-chain data modifications** while maintaining privacy and efficiency. This ensures that **machine learning predictions and user-specific transformations** remain **traceable without exposing raw data**.

- **Developing a Common Ag-Blockchain API**
  Design a **scalable blockchain API** to enable **data transformation and interoperability** across **Farm Management, Food Traceability, and Sustainability applications**. This will facilitate **seamless integration between diverse agricultural systems, improving blockchain adoption and scalability**.

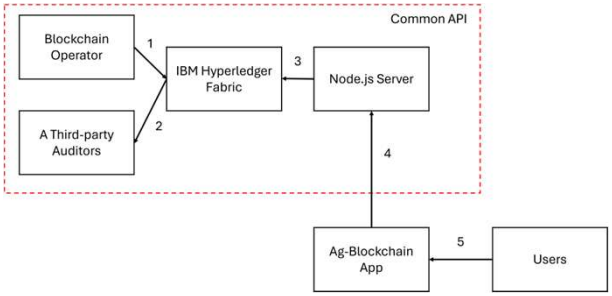



Fig. 3. Proposed framework for scalable and auditable agricultural blockchain system.